

“SEVERAL MINUTES... BEFORE **TOTALITY**, I... FOUND THAT THE **SUN'S IMAGE** MAY BE VIEWED WITHOUT THE SLIGHTEST INCONVENIENCE.”

Warren de la Rue, British Astronomer

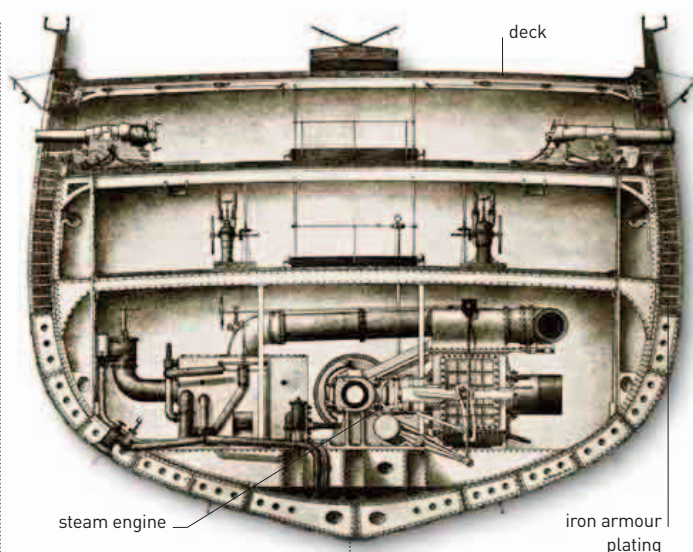


Warren de la Rue developed a special camera that allowed him to photograph a total eclipse of the sun as seen from Spain on July 18, 1860.

THE STIR CAUSED BY THE PUBLICATION OF DARWIN'S

Origin of Species came to a head in fierce public debate in June at the Oxford House Museum in Oxford, UK. Opposing Darwin from a religious perspective was Bishop **Samuel Wilberforce** (1805–73); supporting Darwin and science was the British biologist **Thomas Huxley** (1825–95). The **debate focused on whether humans are descended from apes**, even though Darwin had not actually suggested they were. Huxley is generally considered to have won the debate.

Following their breakthrough in 1859, Bunsen and Kirchhoff made further progress in the field of spectroscopy. Bunsen **discovered two new elements from the spectra of light** absorbed by drops of mineral



water. Each was named after the brightest colors in their spectra: cesium, from the Latin for “sky blue,” and rubidium, meaning “dark red.” Kirchhoff learned more about the sun’s composition from the spectrum

of sunlight, discovering more than 16 different elements in it. **Photographs taken of a solar eclipse** by British astronomer **Warren de la Rue** (1815–89) proved that the flames, now known as prominences, that sometimes appear around the Moon during an eclipse come from the Sun’s surface.

The **oldest known recording of a human voice** was made in April 1860 using a phonautograph made by French bookseller **Édouard-Léon Scott de Martinville** (1817–79) that used sound vibrations to draw on a carbon-coated cylinder. The phonautograph was not designed to play back sound, it was just used for turning sound into a

HMS Warrior

With its armor-plated iron hull, HMS Warrior was the first modern battleship. It was armed with 26 muzzle-loading cannons that were able to fire 68 lb (31 kg) shot over 8,858 ft (2,700 m).

graphic. The recording has only recently been converted to sound using computer technology.

British scientist **Joseph Wilson Swan** (1828–1914) demonstrated the **first working incandescent light bulb**. The light was created by passing an electric current through a thin carbon filament inside a vacuum-filled glass bulb, which heated the filament until it glowed. With only a partial vacuum in the bulb, however, the filament quickly burned out. He made improvements and patented it in 1878 (see 1878–79).

4½
INCHES

THE WIDTH OF HMS WARRIOR'S BELT OF WROUGHT IRON

The launch of HMS *Warrior* on December 29 on the Thames in London was a landmark in naval technology. *Warrior* was only the second **armor-plated, iron-hulled warship** to be built, after the French ship *La Gloire* of 1859, and it was on a vastly different scale from anything that had gone before, at over 417 ft (127 m) long and weighing almost 10,000 tons.

More peacefully, British botanist **Joseph Hooker** (1817–1911) concluded his **account of the many previously unknown plants** he had discovered on his voyages to Antarctica, between 1839 and 1843 aboard the naval ships *Erebus* and *Terror*.

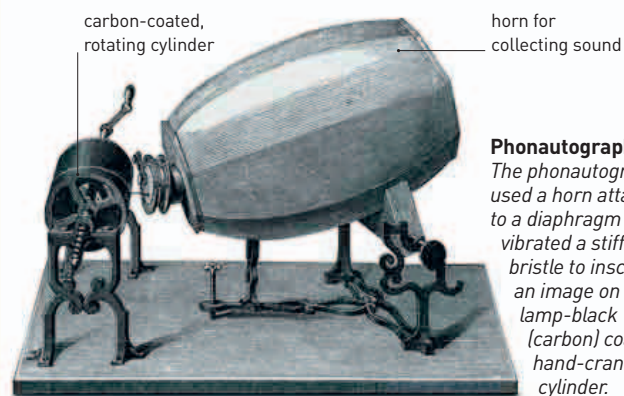
Number of plant species
New species of plants are being identified all the time; 400,000 are now known. Joseph Hooker added 1,500 on his Antarctic voyages.

400,000

THE APPROXIMATE
NUMBER OF KNOWN
PLANT SPECIES IN
THE WORLD

1,500

The number of plant species collected on the Antarctic voyages of Joseph Hooker aboard *Erebus* and *Terror*



Phonautograph

The phonautograph used a horn attached to a diaphragm that vibrated a stiff bristle to inscribe an image on a lamp-black (carbon) coated, hand-cranked cylinder.

German chemist Robert Bunsen **discovers the elements caesium and rubidium** from their spectra

British scientist Joseph Swan demonstrates the **first working incandescent light bulb**

May 26 British Archaeologist Joseph Prestwich presents a paper on his **stone ax** finds to the Royal Society in London

July 9 Nightingale nursing school opened at St. Thomas' Hospital in London

September 3–5 Karlsruhe Congress in Germany, first international meeting of chemists

British botanist Joseph Hooker concludes his **Antarctic discoveries** on the ships *Erebus* and *Terror*

April 9 French bookseller Édouard-Léon de Martinville makes the **oldest known voice recording** on his phonautograph

June 30 Major **debate over evolution** at Oxford, UK

July 18 British astronomer Warren de la Rue discovers **w prominences** during an eclipse

December 29 The **iron battleship, HMS Warrior**, launched